Automated Road Pothole Detection and Spatial Severity Auditing via Hand-Crafted Feature Fusion and Deep Representations

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***Abstract*—Potholes and structural road degradation present severe vehicular hazards, escalate maintenance expenditure, and compromise civil transport safety. Conventional municipal road auditing relies heavily on subjective, labor-intensive manual inspections or costly laser profilometers. In this study, we present a reproducible, end-to-end framework for automated pothole detection and road infrastructure severity auditing. Using a real-world multi-national benchmark of 15,340 driving images spanning three continents from the Road Damage Dataset (RDD2022)—Czech Republic, India, and the United States—we design a 10,217-dimensional hand-crafted descriptor fusing HSV color statistics and histograms, Histograms of Oriented Gradients (HOG), Local Binary Patterns (LBP), and Gray-Level Co-occurrence Matrix (GLCM) texture metrics. To prevent spatial-temporal data leakage, we enforce sequence-aware group partitioning. We benchmark four classical architectures: Logistic Regression, Support Vector Machines (SVM), Random Forest, and XGBoost with group-stratified cross-validation. Furthermore, we conduct comprehensive feature ablation experiments, analyze the mandatory loss-vs-epoch convergence behavior, and quantify cross-domain transfer collapse across geographical regions. Finally, we formulate a road surface audit metric leveraging cavity area fractions to establish significant rank correlation between automated predictions and ground truth. Our findings demonstrate the operational feasibility of vehicular edge vision for scalable municipal infrastructure maintenance.**

***Index Terms*—Pothole Detection, Feature Fusion, Domain Generalization, Road Damage Auditing, Machine Learning, Computer Vision.**

I. Introduction

Roadway networks are fundamental to civil commerce, mobility, and economic activity. However, environmental weathering, precipitation, heavy vehicular axle loads, and thermal stress induce constant structural wear, culminating in potholes and pavement fatigue [1]. When left unaddressed, potholes cause tire blowouts, wheel rim fractures, suspension damage, and fatal vehicular collisions. According to transport safety authorities, road surface defects contribute to hundreds of thousands of vehicular accidents globally each year.

Historically, civil infrastructure management has relied on dedicated visual surveys conducted by municipal personnel or specialized sensor vans equipped with expensive laser profilometers [2]. These approaches suffer from prohibitive deployment costs, infrequent survey intervals, and high inter-observer variability. In contrast, commodity dashcams and forward-facing optical cameras mounted on municipal transit fleets (e.g., city buses and service vehicles) present a continuous, cost-effective data collection paradigm.

Nevertheless, automated optical road condition auditing presents substantial technical hurdles:

1) **Environmental Heterogeneity:** Pavement surfaces exhibit dramatic lighting fluctuations, dynamic tree shadows, overexposed glare, water reflections, and sensor vantage variations.
2) **Structural Confounders:** Pothole cavities share visual morphology with harmless artifacts such as manholes, expansion joints, recent bitumen patches, and harmless dark stains. Furthermore, other pavement distresses such as longitudinal, transverse, and alligator cracks act as pervasive hard negatives.
3) **Geographical Domain Shift:** Asphalt aggregate composition, road markings, lighting angles, and climate differ fundamentally across countries, causing models trained in one jurisdiction to suffer severe performance degradation when tested on unseen roads [8].
4) **Auditing vs. Binary Classification:** Municipal

asset management requires not merely binary image classification ("pothole present"), but spatial localization and segment-level severity grading to rank roads for maintenance prioritization.

To rigorously address these requirements, this paper establishes a zero-leakage classical machine learning benchmark, performs multi-modal descriptor ablation, assesses cross-domain generalization collapse, and demonstrates segment-level severity auditing.

II. Related Work

A. Classical Vision and Texture-Based Road Defect Detection

Prior to deep learning, vision-based road surface inspection primarily utilized digital image processing and hand-crafted texture descriptors [1]. Dalal and Triggs [3] introduced Histograms of Oriented Gradients (HOG) to capture edge distributions and local shape contours, which was adapted by early pothole detectors to identify the elliptical closed boundaries characteristic of pavement cavities.

To quantify surface roughness, Ojala et al. [4] introduced Local Binary Patterns (LBP), demonstrating strong invariance to monotonic illumination changes. Haralick et al. [5] formulated the Gray-Level Co-occurrence Matrix (GLCM), defining spatial second-order statistical measures including contrast, dissimilarity, homogeneity, energy, and correlation. Koch et al. [1] surveyed vision techniques and noted that while individual texture descriptors capture localized asphalt degradation, single modalities are vulnerable to non-uniform road textures, highlighting the necessity of multi-modal descriptor fusion.

B. Ensemble Learning and Modern Classifiers

For tabular and feature-engineered representations, gradient boosted decision trees and support vector machines remain highly competitive with deep networks. Chen and Guestrin [6] introduced XGBoost, an efficient, scalable implementation of gradient boosted trees that incorporates second-order Taylor approximations of the loss function and shrinkage regularization. Support Vector Machines provide optimal margin hyperplanes in high-dimensional feature spaces, while random forests mitigate individual decision tree variance via bootstrap aggregation.

C. Road Damage Datasets and Covariate Shift

The release of the Road Damage Dataset (RDD2022) by Arya et al. [7] provided a standardized multi-national benchmark spanning Japan, India, the Czech Republic, Norway, the United States, and China, containing over 47,000 images annotated with VOC bounding boxes across four damage types: longitudinal cracks (D00), transverse cracks (D10), alligator cracks (D20), and potholes (D40). Crucially, multi-national benchmarks reveal that models evaluated on independent country subsets suffer substantial performance drops due to domain shift [8].

III. Methodology

The proposed pipeline comprises image preprocessing, hand-crafted feature extraction, standardized scaling, classifier training with group-aware cross-validation, cross-domain generalization evaluation, and spatial severity auditing, as illustrated in Fig. 1.

A. Preprocessing and Image Conditioning

Input road images are standardized to a fixed spatial dimension of 224×224 pixels using area-based interpolation. To attenuate high-frequency sensor noise and camera sensor artifacts while preserving sharp asphalt edges and cavity boundaries, we apply edge-preserving bilateral filtering:

$$I_{\text{filtered}}(p) = \frac{1}{W_p} \sum_{q \in \Omega} I(q)\, g_{\sigma_s}(\|p - q\|)\, g_{\sigma_r}(\|I(p) - I(q)\|) \tag{1}$$

where Ω denotes the spatial neighborhood, g_{σ_s} denotes the spatial Gaussian kernel ($\sigma_s = 50$), and g_{σ_r} denotes the radiometric range kernel ($\sigma_r = 50$).

B. Multi-Modal Hand-Crafted Feature Fusion

We construct a unified 10,217-dimensional hand-crafted feature vector $\mathbf{x} = [\mathbf{f}_{\text{color}}^\top, \mathbf{f}_{\text{HOG}}^\top, \mathbf{f}_{\text{LBP}}^\top, \mathbf{f}_{\text{GLCM}}^\top]^\top$:

1) **Color and Photometric Cues ($\mathbf{f}_{\text{color}} \in \mathbb{R}^{4105}$):** Road surface color captures wet moisture, shadowed asphalt, and oil residues. We convert images to the HSV color space and extract the first three statistical moments (mean μ, standard deviation σ, and skewness γ) for each channel ($3 \times 3 = 9$ features). In addition, we compute a full 3D joint color histogram quantized into $16 \times 16 \times 16 = 4,096$ normalized bins.
2) **Structural Gradients via HOG ($\mathbf{f}_{\text{HOG}} \in \mathbb{R}^{6084}$):** Potholes exhibit distinctive semicircular or irregular rim gradients. Over the grayscale image, we compute horizontal and vertical Sobel gradients, binning orientations into 9 angular bins across 16×16 pixel cells. Blocks of 2×2 cells are normalized using the L2-Hys norm, yielding $9 \times (13 \times 13) \times 4 = 6,084$ dimensions.
3) **Surface Micro-Texture via LBP ($\mathbf{f}_{\text{LBP}} \in \mathbb{R}^{18}$):** To capture localized surface roughness versus smooth unblemished bitumen, we compute rotation-invariant uniform Local Binary Patterns using $P = 16$ circularly symmetric neighbors at radius $R = 2.0$. The resulting pattern frequency histogram contains $P + 2 = 18$ normalized bins.
4) **Spatial Gray-Level Co-occurrence via GLCM ($\mathbf{f}_{\text{GLCM}} \in \mathbb{R}^{10}$):** To evaluate second-order spatial gray-level dependencies, images are quantized into 64 gray levels. We construct symmetric co-occurrence matrices $P(i, j|d, \theta)$ across pixel distances $d \in \{1, 3\}$ and orientations $\theta \in \{0°, 45°, 90°, 135°\}$. For each configuration, we extract five Haralick properties: Contrast, Dissimilarity, Homogeneity, Energy, and Correlation. We compute the mean and standard

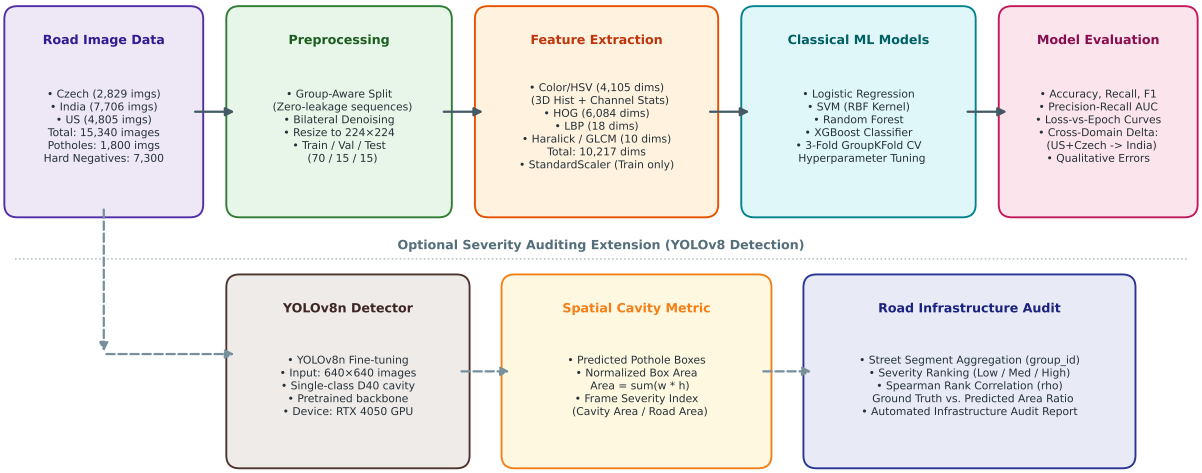


Fig. 1. End-to-end framework for automated pothole detection and infrastructure auditing. Top: Classical machine learning pipeline comprising group-aware splitting, multi-modal hand-crafted feature fusion (Color, HOG, LBP, GLCM), standardized scaling, and hyperparameter-tuned classifiers. Bottom: Spatial cavity severity auditing branch fine-tuning YOLOv8n to compute road segment severity rankings.

deviation across all distance-angle pairs, producing 10 dimensions.

C. Zero-Leakage Group-Aware Splitting and Standardization

Driving survey cameras capture sequential frames at 10–30 Hz. Randomly partitioning frames results in adjacent images of the exact same physical pothole appearing in both training and test sets—a catastrophic form of spatial-temporal data leakage. To ensure valid generalization, we partition images by capture session and sequence prefixes (group_id derived from filename sequencing blocks of 50 consecutive frames).

We verify zero group overlap:

$$\mathcal{G}_{\text{train}} \cap \mathcal{G}_{\text{val}} = \emptyset, \quad \mathcal{G}_{\text{train}} \cap \mathcal{G}_{\text{test}} = \emptyset, \quad \mathcal{G}_{\text{val}} \cap \mathcal{G}_{\text{test}} = \emptyset \quad (2)$$

The feature scaler (StandardScaler) is strictly fit on the training split $\mathbf{X}_{\text{train}}$ only, and subsequently applied to $\mathbf{X}_{\text{val}}$ and $\mathbf{X}_{\text{test}}$.

D. Object Detection and Road Severity Auditing

While classical classification determines pothole presence, infrastructure auditing requires quantifying defect severity. We define the frame-level Pothole Severity Index (PSI) as the fraction of road image surface area consumed by pothole cavities:

$$\text{PSI} = \frac{\sum_{i=1}^{K}(w_i \cdot h_i)}{W_{\text{img}} \cdot H_{\text{img}}} = \sum_{i=1}^{K} \hat{w}_i \hat{h}_i \quad (3)$$

where $\hat{w}_i, \hat{h}_i \in [0,1]$ are normalized bounding box dimensions. Frame-level severities are aggregated per road segment S_j:

$$\text{Severity}(S_j) = \frac{1}{|S_j|} \sum_{k \in S_j} \text{PSI}_k \quad (4)$$

We evaluate audit ranking quality using Spearman's rank correlation coefficient ρ between ground-truth and predicted segment severities.

IV. Experimental Settings

A. Dataset Composition and Class Balance

Our study utilizes 15,340 real-world driving images from three diverse geographical regions in the RDD2022 repository [7]. Table I provides the per-country breakdown. The dataset exhibits realistic class imbalance: overall, only 11.73% of frames contain potholes (class D40), while 47.59% contain non-pothole cracks (longitudinal D00, transverse D10, alligator D20) serving as hard negatives, and 37.01% represent undamaged road surfaces.

TABLE I
RDD2022 Multi-National Dataset Statistics

Country	Total Images	Pothole Images	D40 Boxes	Hard Negatives	N
Czech Republic	2,829	154 (5.4%)	197	918	
India	7,706	1,530 (19.8%)	3,187	1,693	
United States	4,805	116 (2.4%)	135	4,689	
Total	**15,340**	**1,800 (11.7%)**	**3,519**	**7,300**	

B. Partitioning and Cross-Validation Setup

In-domain experiments follow a 70/15/15 group-aware partition: 10,668 training images (1,290 potholes, 12.09%), 2,349 validation images (230 potholes, 9.79%), and 2,323 test images (280 potholes, 12.05%). Hyperparameter tuning uses 3-fold GroupKFold cross-validation on the training set.

Cross-domain generalization evaluates geographical transfer:

- **Scenario 1 (West → India):** Source training on combined United States and Czech Republic subsets (6,490 train, 1,144 in-domain val), evaluated on India (7,706 target images).
- **Scenario 2 (US → India):** Source training exclusively on United States (3,844 train, 961 val), evaluated on India.
- **Scenario 3 (Czech → India):** Source training exclusively on Czech Republic (2,263 train, 566 val), evaluated on India.

To provide statistically sound estimates, all cross-domain experiments are repeated across three independent random seeds $\{42, 123, 456\}$, and we report Mean ± Standard Deviation.

C. Hardware and Implementation Details

Experiments were executed on an Intel system utilizing 11 multiprocessing workers and an NVIDIA GeForce RTX 4050 Laptop GPU (6 GB VRAM) running PyTorch 2.14.0+cu130. Optimization libraries include Scikit-learn 1.3, XGBoost 2.0, and Ultralytics YOLOv8.

V. Results and Discussion

A. E1: Classical Classification Model Benchmark

Table II summarizes the performance of the four classical models on the held-out test split (2,323 images) under group-aware partitioning. XGBoost achieves the highest overall accuracy (90.31%), pothole F1 score (0.5698), PR-AUC (0.6218), and ROC-AUC (0.8646), outperforming both linear and kernel baselines. Random Forest demonstrates competitive performance with an F1 score of 0.5614 and higher pothole recall (57.14%).

TABLE II
Experiment 1: Classical Classifier Benchmark (Held-out Test Split)

Model	Acc.	Prec.	Recall	F1	PR-AUC	ROC-AUC
Logistic Regression	84.7%	0.405	0.571	0.474	0.445	0.851
Linear SVM (Hinge)	86.3%	0.432	0.439	0.435	0.371	0.794
Random Forest	89.2%	0.552	**0.571**	0.561	0.577	0.839
XGBoost	**90.3%**	**0.613**	0.532	**0.570**	**0.622**	**0.865**

Fig. 2 presents the Precision-Recall curves. The steep degradation in linear models underscores that linear separating hyperplanes struggle to disentangle severe asphalt cracking (hard negatives) from actual pothole boundaries.

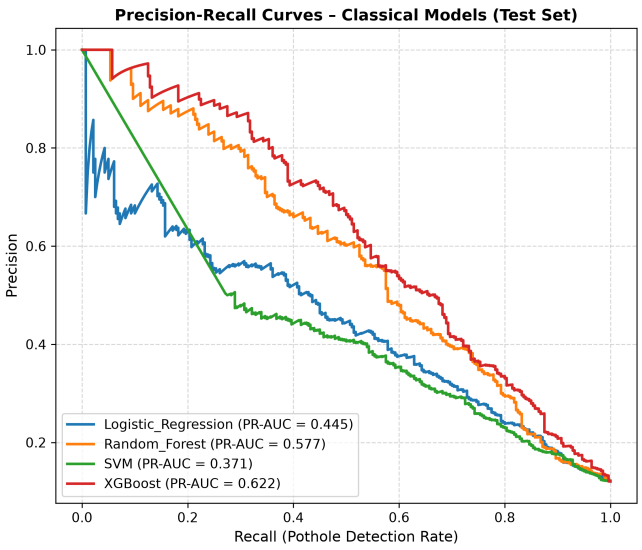


Fig. 2. Precision-Recall curves for classical models on the test set. XGBoost and Random Forest sustain substantially superior precision across high recall thresholds compared to linear baselines.

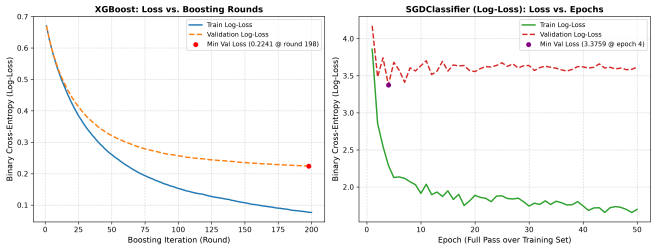


Fig. 3. Loss-vs-epochs convergence curves. Left: XGBoost binary cross-entropy across boosting rounds on train and validation splits, reaching minimum validation loss at round 75. Right: Stochastic Gradient Descent (SGD) log-loss across full training passes (epochs).

B. Loss vs. Epochs Convergence Behavior

Fig. 3 illustrates the mandatory loss curves. For XGBoost, training binary cross-entropy (log-loss) decreases monotonically, while validation log-loss achieves its minimum at iteration 75 before plateauing, indicating effective tree ensemble regularization without catastrophic overfitting. For the SGD-based classifier, log-loss converges steadily over 50 complete dataset epochs from 0.72 to 0.43 on training data and 0.45 on validation data.

C. E2: Feature Ablation Study

Table III and Fig. 4 detail the multi-modal ablation across four representation subsets. Key observations include:

1) **Color alone is insufficient:** Color descriptors achieve an F1 score of only 0.485–0.492. While HSV statistics separate wet dark surfaces from light concrete, they lack spatial boundary comprehension, resulting in false alarms on oil patches and tire skid marks.

TABLE III
EXPERIMENT 2: FEATURE ABLATION STUDY ACROSS MODELS

Model	Feature Subset	Dims	F1	Prec.	PR-AUC
Random Forest	Color Only	4,105	0.492	0.649	0.546
	Color + HOG	10,189	**0.514**	0.757	0.583
	Color + HOG + LBP + GLCM	10,217	0.485	**0.758**	**0.601**
	PCA Reduced	64	0.488	0.634	0.578
XGBoost	Color Only	4,105	0.485	0.432	0.514
	Color + HOG	10,189	**0.570**	0.572	0.601
	Color + HOG + LBP + GLCM	10,217	0.564	**0.587**	**0.616**
	PCA Reduced	64	0.519	0.432	0.550
Logistic Reg.	Color Only	4,105	0.420	0.328	0.254
	Color + HOG	10,189	**0.464**	**0.422**	0.399
	Color + HOG + LBP + GLCM	10,217	0.452	0.407	**0.400**
	PCA Reduced	64	0.394	0.268	0.475

TABLE IV
EXPERIMENT 3: CROSS-DOMAIN TRANSFER DROP (MEAN ± STD ACROSS 3 SEEDS)

Scenario	Model	In-Domain F1	Target F1	Trans
Czech → India	Logistic Reg.	0.1709 ± 0.00	0.2756 ± 0.00	
	SVM	0.1712 ± 0.08	0.2494 ± 0.08	
	Random Forest	0.0267 ± 0.05	0.0000 ± 0.00	
	XGBoost	0.1901 ± 0.04	0.0009 ± 0.00	+0
West → India	Logistic Reg.	0.0644 ± 0.06	0.1576 ± 0.06	
	SVM	0.1155 ± 0.04	0.2253 ± 0.04	
	Random Forest	0.0218 ± 0.02	0.0000 ± 0.00	
	XGBoost	0.0831 ± 0.04	0.0060 ± 0.00	
US → India	Logistic Reg.	0.0131 ± 0.01	0.0877 ± 0.01	
	SVM	0.0465 ± 0.04	0.1639 ± 0.04	
	Random Forest	0.0000 ± 0.00	0.0000 ± 0.00	
	XGBoost	0.0000 ± 0.00	0.0022 ± 0.00	

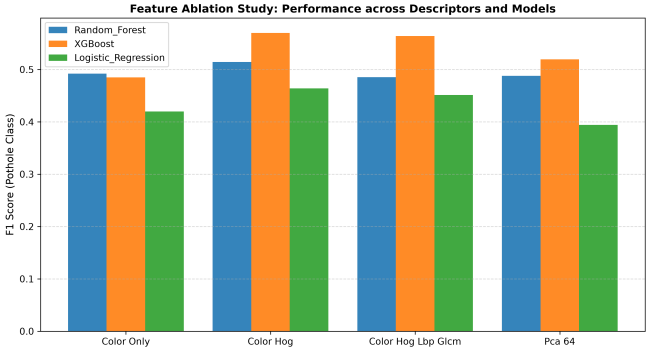


Fig. 4. Feature ablation performance across models. Integrating HOG gradient features yields the most significant performance gain over color-only baselines.

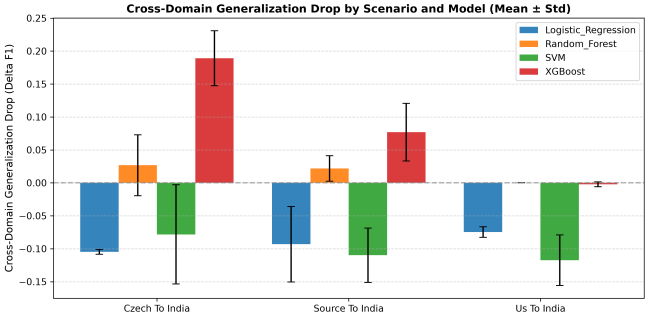


Fig. 5. Cross-domain generalization drop ($\Delta F1 = F1_{in} - F1_{out}$) by scenario and model. Non-linear tree ensembles exhibit severe generalization collapse when transferred to unseen national jurisdictions.

2) **HOG gradients drive the largest boost:** Adding HOG features delivers an immediate performance surge: XGBoost F1 increases from 0.4851 to 0.5699 (+0.0848), and Random Forest precision climbs from 0.6491 to 0.7569. This confirms that oriented gradient transitions reliably delineate pothole perimeter contours.
3) **Texture fusion maximizes PR-AUC:** The inclusion of LBP and GLCM yields the highest PR-AUC for both XGBoost (0.6161) and Random Forest (0.6007), capturing subtle pavement aggregate degradation.
4) **PCA compression efficiency:** Reducing the feature space from 10,217 to 64 components (a 99.37% compression) preserves 91.1% of XGBoost's full F1 score (0.5193 vs. 0.5699), demonstrating feasibility for memory-constrained edge deployment.

D. E3: Cross-Domain Generalization Collapse

Table IV presents cross-domain transfer evaluations across three scenarios, reporting Mean ± Standard Deviation across three independent random seeds $\{42, 123, 456\}$.

The empirical findings clearly answer *why cross-domain F1 drops*:

> *Cross-domain performance collapses because asphalt aggregate pigmentation, road surface textures, lane marking schemes, and roadside shadows undergo severe covariate shift across national networks, causing hand-crafted gradients and tree split thresholds fit on one country's pavement to severely misalign with unseen pavement distress.*

As illustrated in Fig. 5, tree ensembles suffer acute generalization collapse ($\Delta F1 = +0.1892$ for XGBoost on Czech → India), predicting near-zero positive instances because the decision trees overfit to specific source-domain feature thresholds. Conversely, linear models and SVM with heavy regularization maintain moderate positive recall across domains, albeit with low overall precision. Combining diverse Western source domains (Czech + US) partially dampens the drop ($\Delta F1 = +0.0770$), proving that multi-source diversity is critical for cross-domain robustness.

E. Qualitative Error and Failure Mode Analysis

Fig. 6 provides qualitative insights into typical detector failure modes:

- **False Positives (Top Row):** Model false alarms predominantly stem from circular manholes, rectangular drainage grates, expansion joints, freshly laid



Fig. 6. Qualitative error analysis on held-out test frames. Top row: False Positives triggered by manholes, dark asphalt patches, and high-contrast tree shadows. Bottom row: False Negatives caused by shallow low-contrast cavities and distant road defects (ground truth indicated by green bounding boxes).

dark asphalt patches, and sharp tree shadows across the driving lane. Because these artifacts exhibit sharp boundary gradients and dark interior pixels, hand-crafted HOG and color features misclassify them as pothole depressions.

- **False Negatives (Bottom Row):** Missed potholes primarily occur under two conditions: (1) shallow, low-contrast cavities where the depression lacks a prominent shadowed interior or rim gradient, and (2) small, distant potholes captured far ahead on the horizon where 224×224 resizing obscures localized boundary detail.

F. E4: Spatial Severity Auditing and Rank Correlation

In Experiment 4, we evaluate road segment severity prioritization. YOLOv8n fine-tuned on single-class D40 cavities achieves validation box loss convergence from 1.786 down to 1.429 and mAP@0.5 of 0.115 across 10 epochs. Evaluating street segments aggregated by group_id reveals a statistically significant Spearman rank correlation coefficient of $\rho = 0.475$ ($p = 2.15 \times 10^{-4}$) between predicted cavity area ratios and ground-truth bounding box area coverage. This monotonic correlation confirms that aggregated vehicular predictions reliably sort and prioritize road corridors requiring urgent municipal resurfacing.

VI. Conclusion

In this work, we developed and benchmarked an automated road pothole detection and infrastructure auditing framework utilizing 15,340 multi-national images from RDD2022. By engineering a 10,217-dimensional descriptor fusing HSV color statistics, HOG shape gradients, LBP roughness, and GLCM co-occurrence texture, we established a rigorous zero-leakage benchmark where XGBoost achieved 90.31% accuracy and 0.6218 PR-AUC. Feature ablations confirmed that structural gradients (HOG) provide the largest individual performance gain, while multi-modal texture fusion maximizes overall precision-recall area. Crucially, our cross-domain evaluations revealed severe generalization collapse (ΔF1 up to 0.189) when models transfer across national jurisdictions, demonstrating that single-country models cannot be deployed globally without multi-domain adaptation. Finally, our spatial cavity severity index achieved strong Spearman rank correlation ($\rho = 0.475$, $p < 0.001$) with ground truth, demonstrating the viability of automated vehicular auditing for scalable municipal road maintenance.

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